

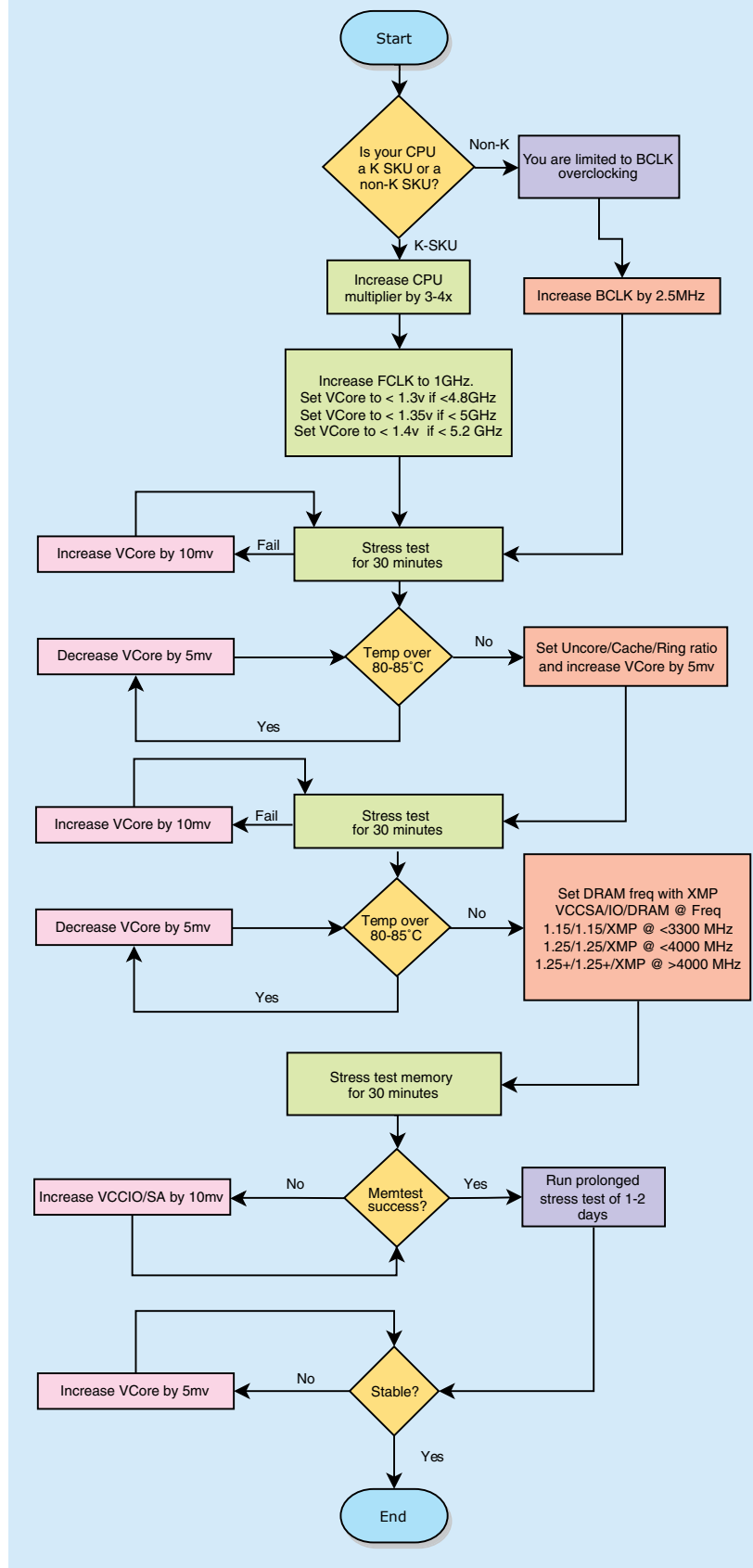
marketed for that purpose and are really expensive. However, you can make do with even the cheapest board with the right chipset. It's just that lower-end boards will not have a good VRM circuit to achieve the higher overclock values. In a similar fashion, don't simply buy the most expensive board from any brand. Evaluate the BIOS features and parameters that you can tweak. Generally the top 3-5 overclocking motherboards will do the job if you want to break records. For everybody else, the mid-range boards form the sweet spot.

The right PSU

The third pillar of overclocking is the Power Supply. When you overclock, the power consumed by the CPU exceeds 140-150 Watts. The motherboard VRM does the job of providing a stable power supply to the CPU but if the PSU that the VRM is drawing from isn't up to the job, then the VRM will be affected as well. Use a power supply calculator such as the one on outervision.com to understand your power requirements and then get a PSU that's fit for the job.

Cooler

The last and equally important component is the CPU cooler which not only has to draw heat from the CPU but also dissipate it quick enough when the processor is under heavy load. Your run of the mill heatsinks are designed for normal usage when the CPU is operating within expected parameters. Overclocking your CPU can raise temperatures by a lot and if the cooling is insufficient, then the CPU's protective mechanisms will throttle it. It's not that overclocking requires an AIO or a custom water cooling loop, you can make do with air coolers such as the Noctua NH-D15 as well. Only certain CPUs like AMD's threadripper will have limited choices owing to the massive package size. Liquid coolers are better off because the cheapest AIO will mostly outperform all mid-range air coolers. Anything with powerful fans and a 360mm radiator should be sufficient for moderate overclocks. For extreme overclocks, it's better to go with a copper pot for LN2 cooling.



Know your ranges

Every CPU generation and even the SKU have safe ranges for the voltage. It's always advisable to keep your values within this range to ensure that your CPU does not burn out. Most current gen (6th-8th) Intel CPUs do well if

the core voltage (VCore) is kept within the 1.3 to 1.4v range whereas AMD recommends values between 1.35 to 1.45v for the new Ryzen CPUs. There are also reference multiplier values that you should know for your CPU. For example, the new Intel unlocked